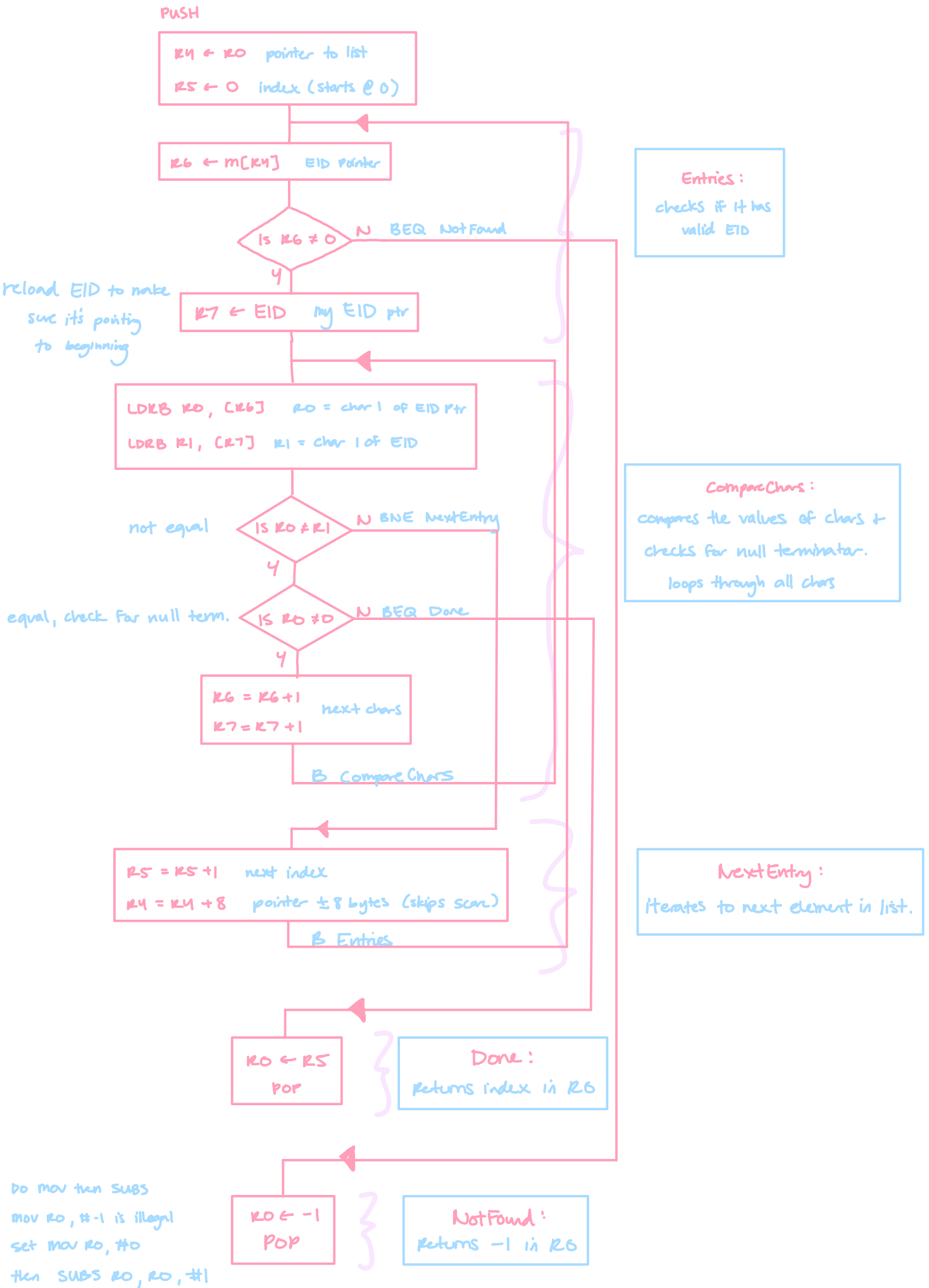
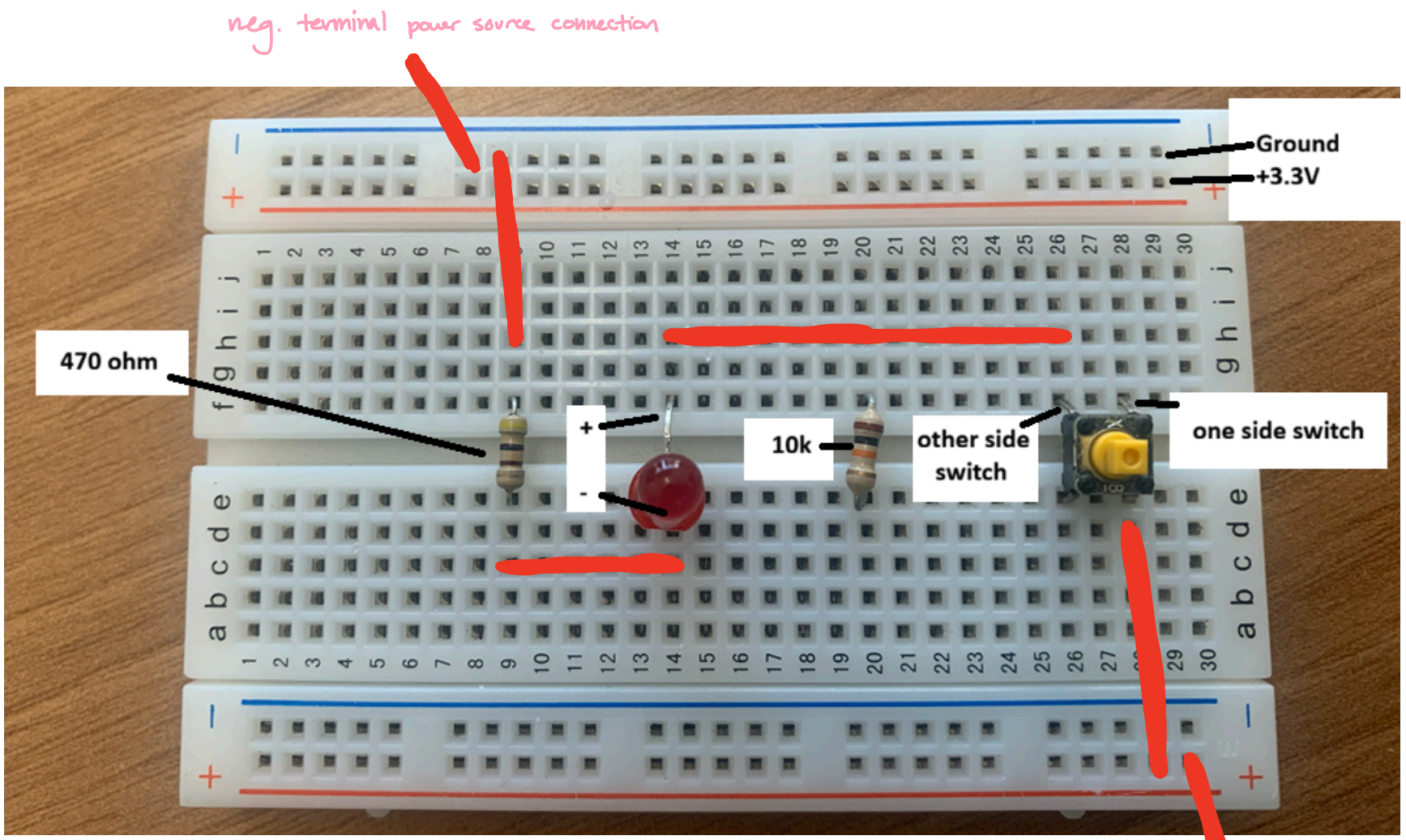

PUSH
R4 ← R0 pointer to list
R5 ← 0 index (starts @ 0)
R6 ← m[R4] EID pointer
Is R6 ≠ 0
N BEQ NotFound
Y
Reload EID to make sure it's pointing to beginning
R7 ← EID my EID ptr
LDRB R0, [R6] R0 = char 1 of EID ptr
LDRB R1, [R7] R1 = char 1 of EID
not equal
IS R0 ≠ R1
N BNE NextEntry
Y
equal, check for null term.
IS R0 ≠0
N BEQ Done
Y
R6 = R6 +1
R7 = R7 +1
next chars
B CompareChars
R5 = R5 +1 next index
R4 = R4 +8 pointer ±8 bytes (skips scan)
B Entries
R0 ← R5
POP
Do MOV then SUBS
MOV R0, #-1 is illegal
set MOV R0, #0
then SUBS R0, R0, #1
R0 ← -1
POP
Entries :
checks if it has valid EID
CompareChars :
compares the values of chars + checks for null terminator.
loops through all chars
NextEntry :
iterates to next element in list.
Done :
Returns index in R6
NotFound :
Returns -1 in R6


neg. terminal power source connection
Ground
+3.3V
470 ohm
+
-
10k
other side switch
one side switch
pos. terminal power source connection